3.1 Data Preparation

During this step, the data will be massaged and transformed into a format that would be compatible with each other for analysis. The crime and GDP data are in their raw format as per below:

crime_district.csv			
district	year	state	crimes
Alor Gajah	2015	Melaka	424
Alor Gajah	2016	Melaka	436
Alor Gajah	2017	Melaka	376

gdp_district_real_supply.csv					
series	state	district	date	sector	value
abs	Johor	Batu Pahat	1/1/2015	p0	11605.5
abs	Johor	Batu Pahat	1/1/2015	p1	2177.257
abs	Johor	Batu Pahat	1/1/2015	p2	13.0113

Table 3.1: Sample Data before Data Preparation

Seeing as this data is not formatted in a district-year-variable format, it needs to be massaged and transformed (on both sides) into this format before they can be merged. This requires data deduplication to remove crime categories from the crime data and data massaging to pivot rows to columns for the GDP data on the right.

The data will then be merged and transformed using some of the sample code attached in Appendix. The result of the transformation (pivoting, duplicate removal, missing value treatment, merging etc.). Once the datasets are merged, some final preparation is needed before it is ready for modelling. In the state it is in Table 3.2, the data would only be useful to predict the crimes of a district in the same year the GDP is made available. In terms of practical applications, this is not very helpful. Hence, the feature columns of the dataset need to be lagged by a year (shifted one row up) for each district. The code to do this is also included in the Appendix and the resulting dataset sample is shown in Table 3.2.

year	district	crimes	gdp_purchasers_l1	gdp_agriculture_l1	gdp_manufacturing_l1	gdp_services_l1
2015	Alor Gajah	424	1477	726.8712	10.7435	729.7303
2015	Bachok	364	1395	434.7939	194.1093	719.3754
2015	Baling	376	600	265.7816	35.6816	286.8733

Table 3.2: Sample Data after Data Preparation

3.2 Supervised Learning (Regression)

The target variable is crime while the features are gdp_purchasers_l1, gdp_agriculture_l1, gdp_manufacturing_l1, and gdp_services_l1 which represent the respective GDP values of that district and in the year before. District and year are excluded from the machine learning model because there are insufficient examples of each district, to allow the model to learn (district has a very high cardinality and cannot be encoded). The year is not deemed a relevant variable in this problem area.

First, correlation analysis will be done to find the variables that have a correlation to the target variable. Based on the correlation analysis, decisions can be made on the features that can be selected and excluded. Then, the regression models will be trained in separate experiments involving 5 models of different varieties. The best model that has a good trade off of performance without any concessions will be selected as the best model.

3.3 Unsupervised Learning (Clustering)

An unsupervised learning approach will also be used to identify clusters of crime and GDP. The same dataset prepared for regression will be used for clustering. All variables (both crime and GDP variables) will be used as features. The algorithms to be used are KMeans clustering, Agglomerative clustering, and DBScan.

The data will first need to be scaled for all features to be on the same range, and to ensure that they are all well-distributed throughout the scale. The code to perform scaling is included in the Appendix. Then, the Elbow method is used to determine the suitable number of clusters. For KMeans clustering, the Elbow method for inertia will be used. For Agglomerative clustering, the dendogram will be used. Finally, for DBScan, the Elbow Method with Epsilon will be plotted. Then, the silhouette score will be evaluated and the best performing model with a meaningful number of clusters will be selected as the model to proceed with which can be interpreted for insights. Insights such as different blends and patterns of crime and GDP that are associated with one another could be determined, for example – which are the crime rate thresholds for varying levels of crime (very high, high, low, very low), and what are the associated GDP of these high crime rate district.

4 Results

4.1 Regression

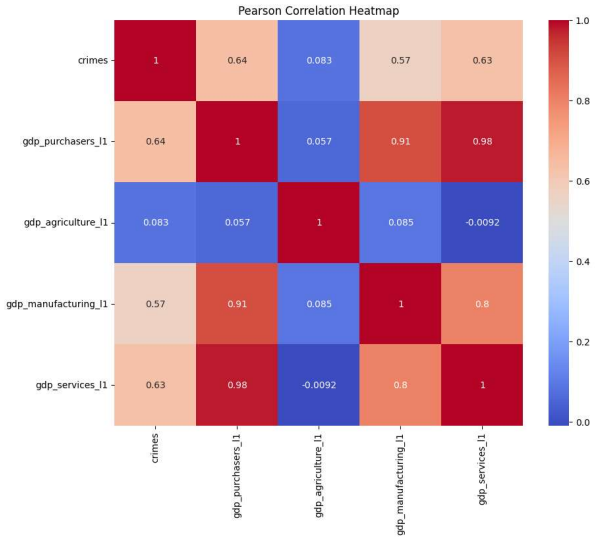


Figure 4.1: Correlation of crimes to gdp features

The correlation analysis in Figure 4.2 shows that there is a high correlation in the GDP values to crime for all GDP sectors except for gdp_agriculture. This intuitively makes sense because one would think that agriculture of a district doing good or bad does not have any bearing on crimes.

There is also a high correlation between the features. We found that this might cause a multicollinearity problem in the model but as we will see later, the model is performing well even with this multicollinearity (with an R2 of more than 80%) and hence, it is safe to assume that this is not a problem.

Based on these correlation analysis insights, we made the decision to implement 2 versions of models i.e. one version with all features, and one version with all features excluding agriculture. The table summarises the performance of the two versions of the model, for all 5 model types that were tried.

Version	Model	MSE	R2	CV Score	Comments
All Features	KNN Regressor	0.183590	0.811343	0.743294	Best non-overfitted model
	Random Forest Regressor	0.130680	0.865713	0.732096	Best R2, but overfitted
	SVR	0.151774	0.844037	0.664049	Overfitted model
	ElasticNet	0.844439	0.132254	0.118033	Worst model in every way
Without agriculture	KNN Regressor	0.241139	0.752206	0.718310	Best non-overfitted model
	Random Forest Regressor	0.233957	0.759586	0.688725	Best R2, but overfitted
	SVR	0.307226	0.684294	0.639601	Overfitted model
	ElasticNet	0.844439	0.132254	0.118033	Worst model in every way

Table 4.1: Summary of Regression Results

As shown in Table 4.4, the results were compared, and it was found that the performance of the model (measured by R2) does not significantly drop by removing agriculture beyond a confidence level of 95%. Considering the correlation analysis and the model results, we decided to remove agriculture from the final model. This is because despite the model performing slightly worse (but not significantly worse), the agriculture feature might be introducing noise to the model and not helping it learn the crimes.

KNN Regressor is the superior model for both versions hence, it is selected as the model to be used for future predictions. This model is ideal not just because it performs well but also because the size of the data is not that big i.e. only about 1000 rows. KNN Regressor struggles with data of large size and can be computationally expensive to run on large data. This is not a problem for our use case because the data is relatively small i.e. only 1000 rows, and this data will not grow much beyond this point because it's size only increases by about 150 news for every year (there are only about 150 districts in Malaysia). This could be looked at again in the future when scaling to other countries is considered but for now, it is deemed fine to opt for KNN regression.

4.2 Clustering

First, the Elbow Method based on inertia measurement resulting from different algorithms yielded the following results:

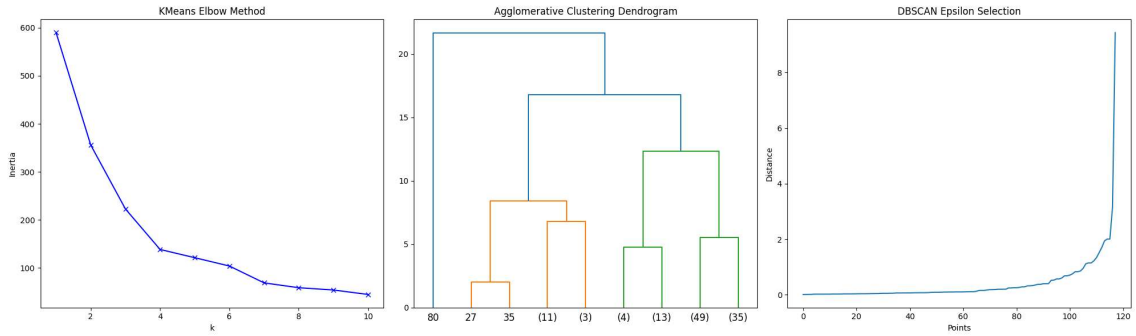


Figure 4.2: Ideal Number of Clusters for Unsupervised Learning

For KMeans clustering, the ideal cluster number is selected as 4, while for Agglomerative clustering, the ideal clusters found are 4 clusters as well. For DBScan, the ideal epsilon is about 2. After setting this hyperparameter for the models, the performance in terms of silhouette score was measured. It was found that Agglomerative clustering performed the best in terms of the clustering models.

Model	Silhouette Score	Comments
DBScan	0.779413	An epsilon of 2 means clusters of noise and not noise. Not meaningful.
Agglomerative	0.546694	The next best model. With 4 clusters, it is meaningful.
KMeans	0.539633	The worst model.

Table 4.2: Clustering Results Summary

As such, the agglomerative clustering model was then used to assign the clusters for each district and year and then the clusters were interpreted for descriptive statistics and similarities that could explain the clusters.

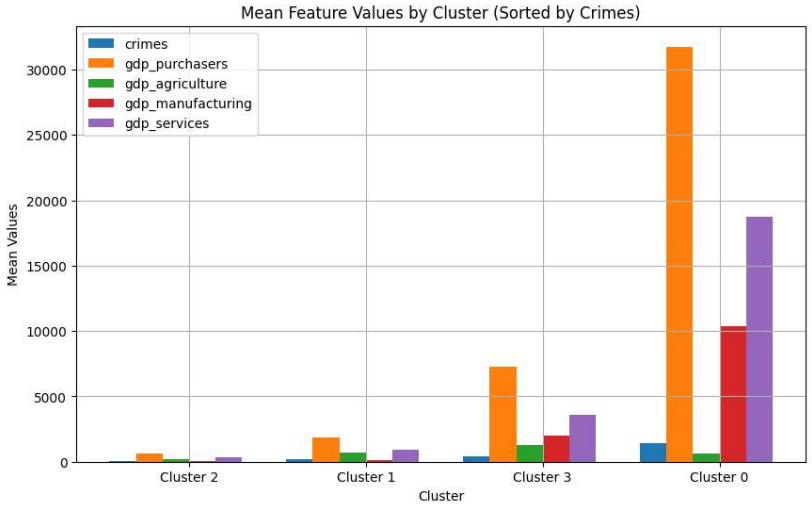


Figure 4.3: Clusters Interpretation for Agglomerative Clustering Model

Based on the insights, it was found that an average crimes of 68.82, 170.61, 413.27, and 1396.02 are very low, low, high, and very high number of crimes for a district. These clusters of crime also show a trend of increasing gdp_purchasers which indicates that high crime clusters also have high gdp_purchasers. Hence, it might be better for the government of Malaysia to place crime fighting efforts in places with gdp_purchasers.

5 Discussion

The findings from both our regression and clustering analyses reveal several significant patterns in the relationship between GDP and crime rates across Malaysian districts. These results have important implications for both policymaking and our understanding of socioeconomic factors in crime prediction.

5.1 Predictive Power of Economic Indicators

The strong performance of our KNN regression model ($R^2 \approx 0.75$) demonstrates that district-level GDP components serve as reliable predictors of crime rates. The exclusion of agricultural GDP without significant performance degradation suggests that crime patterns are more closely tied to urban and industrial economic activities. This aligns with classical criminological theories that link crime to urbanization and industrialization processes. The high predictive power of manufacturing and services GDP particularly suggests that areas with concentrated economic activity may require proportionally increased security measures.

The superior performance of the KNN regressor over other models, including more complex ones like Random Forest, indicates that crime-GDP relationships may be fundamentally local in nature. This locality principle suggests that crime rates in a district are more similar to those in economically similar districts rather than following complex global patterns. This insight could be particularly valuable for police forces in newly industrializing areas, allowing them to learn from the experiences of economically similar districts.

5.2 Economic Development and Crime Clusters

Our clustering analysis revealed a striking pattern where higher crime clusters consistently corresponded with higher GDP levels. This finding challenges simplistic assumptions that economic development automatically leads to crime reduction. Instead, it suggests a more complex relationship where economic growth might create:

1. More criminal opportunities through increased commercial activity
2. Greater wealth disparities within districts
3. Higher population density and urbanization, which may facilitate certain types of crime
4. Increased reporting rates due to better institutional infrastructure
5. More complex social interactions and potential conflict points

The identification of four distinct crime-rate clusters (very low: ~69, low: ~171, high: ~413, very high: ~1,396) provides valuable benchmarks for policy makers to categorize and monitor district-level crime patterns. This clustering could help in resource allocation and in setting appropriate intervention thresholds. The substantial gap between the 'high' and 'very high' clusters (413 vs 1,396) suggests a possible tipping point in crime dynamics that warrants further investigation.

5.3 Policy Implications

Our findings suggest that economic development strategies should be accompanied by proportional investment in crime prevention. Districts experiencing rapid GDP growth, particularly in manufacturing and services sectors, may need preemptive strengthening of law enforcement resources. The strong predictive power of our model also suggests potential applications in:

- Early warning systems for crime rate changes based on economic indicators
- Resource allocation planning for law enforcement
- Development of targeted crime prevention strategies for high-GDP districts
- Integration of crime prevention into economic development planning
- Cross-district learning and best practice sharing based on economic similarities

The clear relationship between economic indicators and crime rates suggests that law enforcement agencies should establish stronger partnerships with economic planning bodies. This could involve:

1. Joint planning sessions for new industrial or commercial developments

2. Shared data systems for monitoring both economic and crime indicators

3. Collaborative risk assessment frameworks

4. Integrated response protocols for rapidly developing areas

5.4 Theoretical Implications and Model Interpretation

The success of our relatively simple KNN model in predicting crime rates has interesting theoretical implications. The fact that a distance-based algorithm performs well suggests that the relationship between economic factors and crime follows relatively stable patterns across similar districts. This challenges more complex theoretical models that posit highly nonlinear or chaotic relationships between economic development and crime.

The removal of agricultural GDP from the model without significant performance loss also provides interesting theoretical insights. This could indicate that:

1. Agricultural economic activity has fundamentally different social dynamics than manufacturing or services

2. The social organization of agricultural communities might provide natural crime prevention mechanisms

3. The relationship between economic activity and crime might be mediated by factors more prevalent in urban and industrial settings

5.5 Limitations and Future Research

Several limitations of our study should be acknowledged:

1. The annual nature of our data may mask shorter-term fluctuations in both crime rates and GDP
2. District-level aggregation might obscure more granular patterns at the neighborhood level
3. The model doesn't account for specific types of crimes, which might have different relationships with economic indicators
4. Potential reporting biases across districts with different economic development levels
5. The absence of social and demographic control variables
6. Limited ability to establish causal relationships despite strong correlations

Future research could address these limitations by:

- Incorporating more granular temporal and spatial data
- Disaggregating crime types to understand specific crime-economy relationships
- Including additional socioeconomic variables such as income inequality, unemployment rates, and demographic factors
- Expanding the analysis to include inter-district effects and spatial relationships
- Conducting comparative studies with other rapidly developing economies
- Investigating the mechanisms behind the agriculture-crime relationship
- Exploring the role of informal economies in crime patterns

5.6 Methodological Contributions

Our study demonstrates the effectiveness of combining supervised and unsupervised learning approaches in criminological research. The complementary insights from regression and clustering analyses provide a more nuanced understanding than either method alone could offer. The success of the KNN regressor, despite its known limitations with large datasets, suggests it may be particularly suitable for district-level analysis in countries with administrative divisions like Malaysia's.

The methodology developed in this study offers several innovations:

1. Integration of economic sector-specific GDP data in crime prediction
2. Development of a robust data preparation pipeline for combining crime and economic data
3. Demonstration of effective feature selection through correlation analysis
4. Validation of simple, interpretable models for complex socioeconomic phenomena